

SCHADN

Intelligent Candidate Discovery

& Ranking System

India Runs
Data & AI Challenge

EXECUTIVE SUBMISSION

Hybrid Offline Ranker for Redrob AI

Semantic fit over keywords · Trap-aware · Production-scale CPU ranking

**Team
Schadn**

**Team Leader
Rahul Kumar Singh**

Official bundle: candidates.jsonl (100K) · job_description.docx ·
redrob_signals_doc.docx · submission_spec v4

Solution Overview

A consultant-grade ranking engine built for real recruiting constraints

Proposed Solution

Rank top 100 from official candidates.jsonl per submission_spec v4.

Reads candidate_schema: profile, career_history, skills, redrob_signals (23 fields).

Hybrid offline ranker: ~49-60s on CPU — within 5 min / 16GB / no-network limits.

Grounded 2-sentence reasoning per row aligned to Stage 4 review criteria.

Strategic Differentiation

Defends bundle traps: keyword stuffers, behavioral twins, ~80 honeypots.

Career semantics + IR skill trust + Redrob behavioral modifier.

Mirrors JD disqualifiers: research-only, consulting-only, CV/speech-only.

open_to_work_flag=false cannot reach top-10.

Shipper-first: scales to 200K+ without per-candidate LLM cost.

JD Understanding & Candidate Signals

Grounded in job_description.docx · candidate_schema.json · redrob_signals_doc.docx

From job_description.docx

Senior AI Engineer — Founding Team @ Redrob AI (Series A).

Pune/Noida preferred · 5–9 years · production IR over research.

Must-have: embeddings retrieval, vector DB, Python,
NDCG/MRR/MAP eval.

Rejects: research-only, LangChain-only, consulting-only, CV without IR.

candidate_schema.json

profile · career_history · skills · redrob_signals (23 fields).

Skills carry proficiency, endorsements, duration_months as trust signals.

23 Redrob Signals (weighted)

Availability: open_to_work_flag, last_active_date, notice_period_days

Responsiveness: recruiter_response_rate, avg_response_time_hours

Engagement: profile_views, saved_by_recruiters, search_appearance

Quality: skill_assessment_scores, github_activity_score

Trust: verified_email, verified_phone, linkedin_connected

Intent: willing_to_relocate, preferred_work_mode, salary_range

Ranking Methodology

Hybrid v6 — retrieve, score, rank, explain

Pipeline

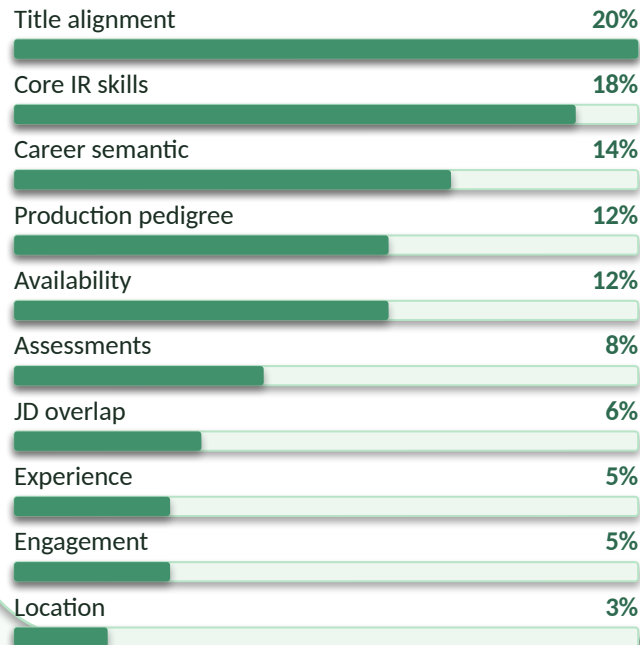
- 1 Parse job_description.docx → jd_config.py + JD embedding
- 2 Load candidates.jsonl (100K) → CandidateIndex
- 3 Score: hybrid v6 + redrob_signals + honeypot traps
- 4 Sort ranks 1-100; tie-break candidate_id ascending
- 5 Export submission.csv — UTF-8, monotonic scores

Models

MiniLM-L6-v2 bi-encoder (embeddings.fp16.npz, 100K vectors)

TF-IDF JD overlap · heuristic modifiers · CE OFF

v6 Signal Weights



Explainability & Data Validation

Transparent decisions judges can audit

Reasoning Quality

2 sentences: specific facts + JD link + honest concerns.

Only cites fields present in profile — zero hallucination.

Avoids: empty, identical, or templated reasoning strings.

Rank tone matches position (strong at #5, candid at #95).

Validated: mock_stage4_review — 10/10 rows, 6/6 checks each.

Trap Defense

~80 honeypots: impossible tenure, expert skills at 0 months.

Keyword stuffers: weak titles + inflated AI skill lists.

Behavioral twins: same profile, different signals — availability wins.

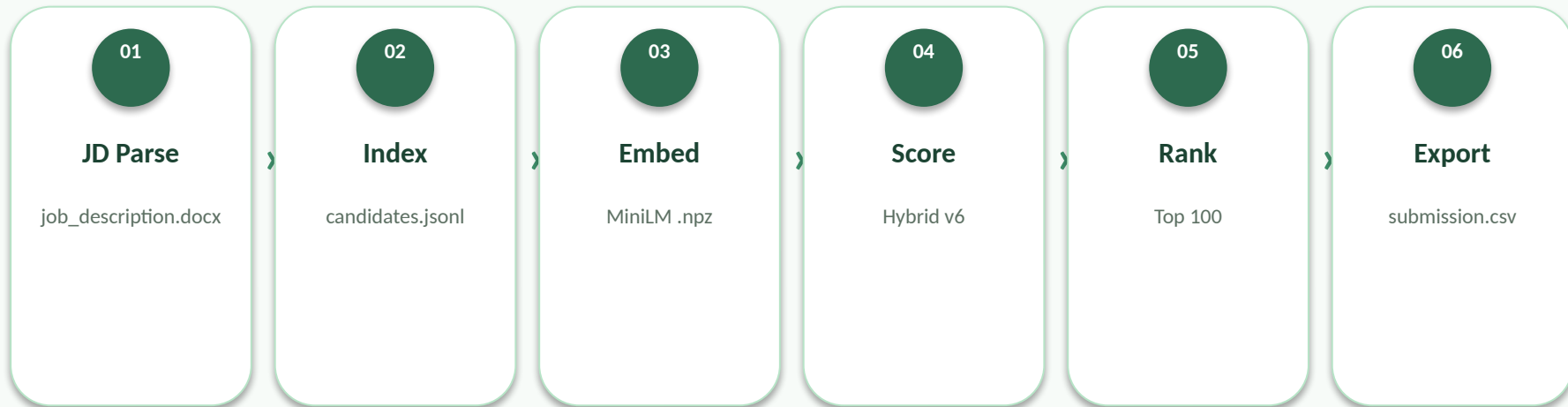
Template blurbs: recycled descriptions demoted.

sample_submission.csv = format only (ranks honeypots by design).

Our result: 0 honeypots in top-100 (limit: 10%).

End-to-End Workflow

From official JD to submission-ready CSV



Reproduce: `./scripts/reproduce_ranking.sh` · Byte-verified via `verify_submission_artifact.py`

System Architecture

Graded submission path vs optional demo platform

GRADED PATH



Input: candidates.jsonl + embeddings.fp16.npz → Output: submission.csv (100 rows)
+ career_blurb.py · assessment.py · rerank.py (cross-encoder OFF)

DEMO PATH (not graded)

Next.js 14 frontend → FastAPI + LangGraph backend → dashboard, analytics, AI chat
Imports top-100 from submission.csv — does not produce the graded artifact.

Results & Performance

Evidence-backed outcomes and spec compliance

Top-5 Finalists

- #1 CAND_0002025 · 0.9900 · Sr AI Engineer @ Apple
- #2 CAND_0046064 · 0.9324 · Sr NLP Engineer @ Salesforce
- #3 CAND_0010685 · 0.8659 · NLP Engineer @ Rephrase.ai
- #4 CAND_0055905 · 0.8510 · Sr ML Engineer @ Flipkart
- #5 CAND_0018499 · 0.8041 · Sr ML Engineer @ Zomato

Validation

100 rows · unique ranks · monotonic scores · 0 honeypots

All IDs verified in official candidates.jsonl

Byte-reproducible artifact: PASS

Reasoning: 100% unique · Stage 4: 10/10 pass

~49–60s

Runtime on 100K
« 5 min limit

CPU

No GPU · No network
Spec compliant

16 GB

RAM envelope
~68MB embeddings

Technologies Used

Deliberate stack choices for reproducibility and scale

Submission Ranker

Python 3.11 · NumPy · PyYAML — no torch at rank time

all-MiniLM-L6-v2 — committed embeddings.fp16.npz

Docker: docker-compose.ranker.yml for Stage 3 reproduction

pytest: 31 tests passing in challenge/test_ranker.py

AI tools: Cursor + Grok (dev only — no candidate data to LLMs)

Demo Platform

Next.js 14 + Tailwind — recruiter command center

FastAPI + LangGraph — 7-agent explainability pipeline

PostgreSQL / SQLite — workspace + top-100 import

HuggingFace Spaces — sample ranking sandbox

Vercel + Render — recruitgpt-x.vercel.app

Submission Assets

Complete reproducibility package for judges

GitHub

github.com/rahulx2001/recruitgpt-x

rank.py + challenge/ + bundle validators

Ranked CSV

submission.csv

submission_spec v4 · UTF-8 · 100 rows

HF Sandbox

huggingface.co/.../recruitgpt-ranker

\$10.5 sample_candidates.json

Official Data

candidates.jsonl + job_description.docx

sync_challenge_data.sh

Reproduce

./scripts/reproduce_ranking.sh

CPU ≤5min · byte-verified

Metadata

submission_metadata.yaml

Team Schadn · Rahul Kumar Singh

Thank You

Schadn · Rahul Kumar Singh

Ranking candidates like a great recruiter — not a keyword filter.

github.com/rahulx2001/recruitgpt-x